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FCFS Scheduling (Non-Preemptive)

Input:

```
1. FCFS Scheduling (Non-Preemptive)
2. Round Robin Scheduling (Preemptive)
3. Banker's Algorithm
4. Exit
Select an option (1-4): 1
Enter number of processes: 3
Enter Arrival Times (space-separated): 0 2 4
Enter Burst Times (space-separated): 1 3 5
```

Output:

```
1. FCFS Scheduling (Non-Preemptive)
2. Round Robin Scheduling (Preemptive)
3. Banker's Algorithm
4. Exit
Select an option (1-4): 1
Enter number of processes: 3
Enter Arrival Times (space-separated): 0 2 4
Enter Burst Times (space-separated): 1 3 5

--- FCFS (Non-Preemptive) Results ---
Process  Arrival  Burst   Waiting  Turnaround
P0       0        1       0         1
P1       2        3       0         3
P2       4        5       1         6

Average Waiting Time: 0.33
Average Turnaround Time: 3.33

Gantt Chart:
| P0 (0-1) | P1 (2-5) | P2 (5-10) |
```

Round Robin (Preemptive)

Input:

```
=====
OPERATING SYSTEM SIMULATOR
=====
1. FCFS Scheduling (Non-Preemptive)
2. Round Robin Scheduling (Preemptive)
3. Banker's Algorithm
4. Exit
Select an option (1-4): 2
Enter number of processes: 3
Enter Arrival Times (space-separated): 3 6 9
Enter Burst Times (space-separated): 4 8 12
Enter Time Quantum: 2
```

Output:

```
=====
OPERATING SYSTEM SIMULATOR
=====
1. FCFS Scheduling (Non-Preemptive)
2. Round Robin Scheduling (Preemptive)
3. Banker's Algorithm
4. Exit
Select an option (1-4): 2
Enter number of processes: 3
Enter Arrival Times (space-separated): 3 6 9
Enter Burst Times (space-separated): 4 8 12
Enter Time Quantum: 2

--- Round Robin (Preemptive) Results ---
Process  Arrival  Burst   Waiting  Turnaround
P0       3        4        0         4
P1       6        8        7        15
P2       9       12        6        18

Average Waiting Time: 4.33
Average Turnaround Time: 12.33

Gantt Chart:
| P0 (3-5) | P0 (5-7) | P1 (7-9) | P2 (9-11) | P1 (11-13) | P2 (13-15) | P1 (15-17) | P2 (17-19) | P1 (19-21) | P2 (21-23) | P2 (23-25) | P2 (25-27) |
```

Banker's Algorithm

Safe State

```
--- Banker's Algorithm Simulation ---
Enter number of processes: 3
Enter number of resource types: 3
Enter Allocation Matrix (row by row, space-separated):
Process 0: 0 1 0
Process 1: 2 0 0
Process 2: 3 0 2
Enter Max Matrix (row by row, space-separated):
Process 0: 7 5 3
Process 1: 3 2 2
Process 2: 9 0 2
Enter Available Resources (space-separated): 3 3 2

Need Matrix:
P0: [7, 4, 3]
P1: [1, 2, 2]
P2: [6, 0, 0]

System is in an Unsafe State.
```

Unsafe State

```
--- Banker's Algorithm Simulation ---
Enter number of processes: 4
Enter number of resource types: 3
Enter Allocation Matrix (row by row, space-separated):
Process 0: 0 1 0
Process 1: 2 0 0
Process 2: 3 0 2
Process 3: 2 1 1
Enter Max Matrix (row by row, space-separated):
Process 0: 7 5 3
Process 1: 3 2 2
Process 2: 9 0 2
Process 3: 4 2 2
Enter Available Resources (space-separated): 3 3 4

Need Matrix:
P0: [7, 4, 3]
P1: [1, 2, 2]
P2: [6, 0, 0]
P3: [2, 1, 1]

System is in a Safe State.
Safe Sequence: P1 -> P3 -> P0 -> P2
```